



Spear3 RF System

- RF Requirement
- Overall System
- High Power Components
- Operation and Control

SPEAR 3 History



1996	Low emittance lattices explored
1996	SPEAR 3 proposed
11/97	SPEAR 3 design study team formed
11/97	Director's Review
07/98	DOE Lehman Review
FY99	DOE BES and NIH discuss joint funding
11/98	Active cavity and WG arcing
01/99	Additional funding for NEW RF (476.3 MHz)
04/99	Active RFHVPS failure.
01/00	Cavities ordered (Received 05/03)
03/00	Klystron ordered (Received 08/01, Repaired 05/03)
05/01	2.5 MW PS ordered (Received 01/02)
11/01	Circulator ordered (Received 11/01)
02/02	WG parts ordered (Receive 04/02)
03/02	LLRF work in progress
04/03	Installation (6 months)
12/03	Commissioning (3 months)
03/04	User Beam (3.0 GeV, 100 mA, 18 nm-rad)

Electron Beam Energy Loss due to Synchrotron Radiation



Energy loss at bend magnets

$$U_{0\text{-bend}} (\text{keV/turn}) = 88.5 \cdot (E_b/\text{GeV})^4 / (\rho/\text{m})$$

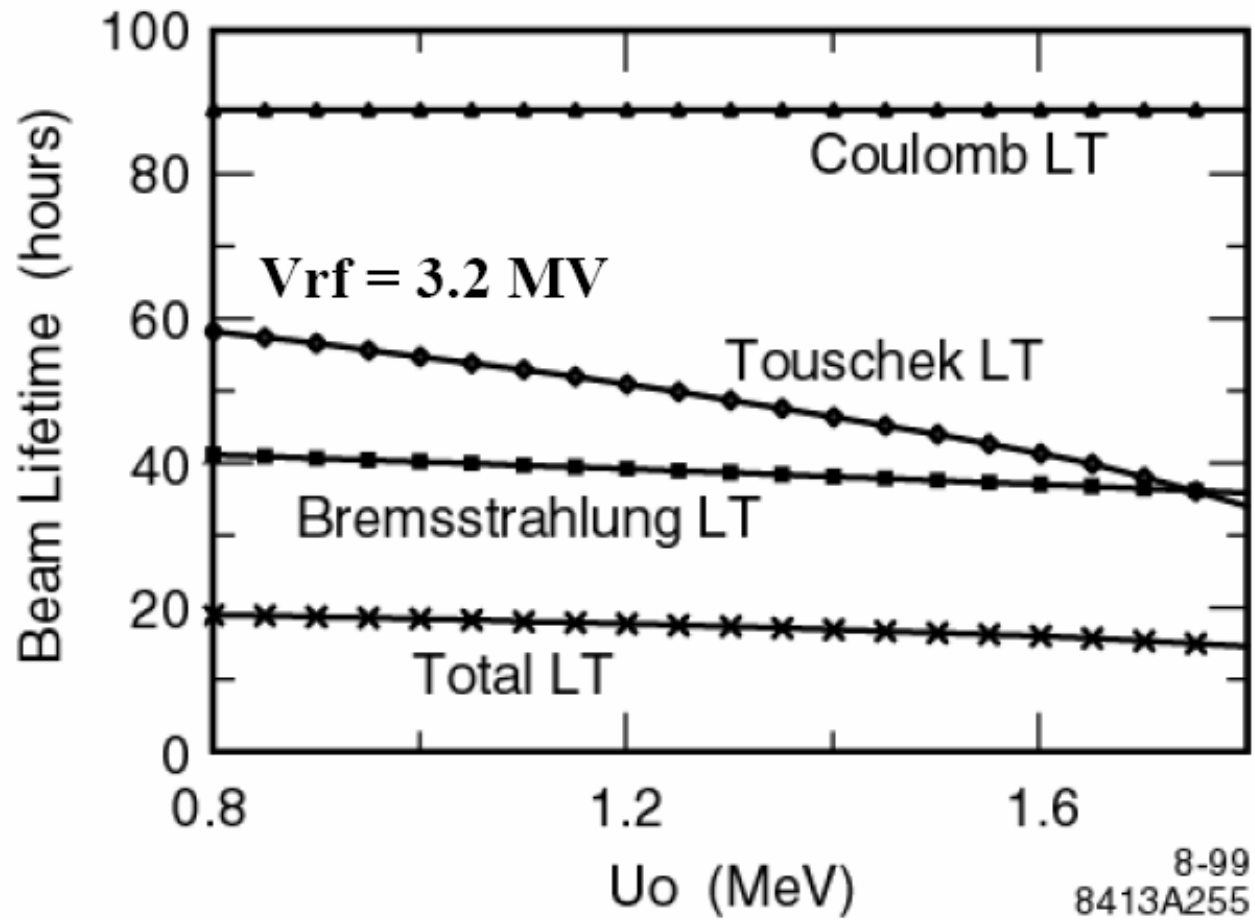
Energy loss at insertion device

$$U_{0\text{-ID}} (\text{keV/turn}) = 0.633 \cdot (E_b/\text{GeV})^2 \cdot \langle (B/T)^2 \rangle \cdot (L/\text{m})^2$$

where $\langle B \rangle$ is the rms magnetic field of the pole
and L is the insertion device length

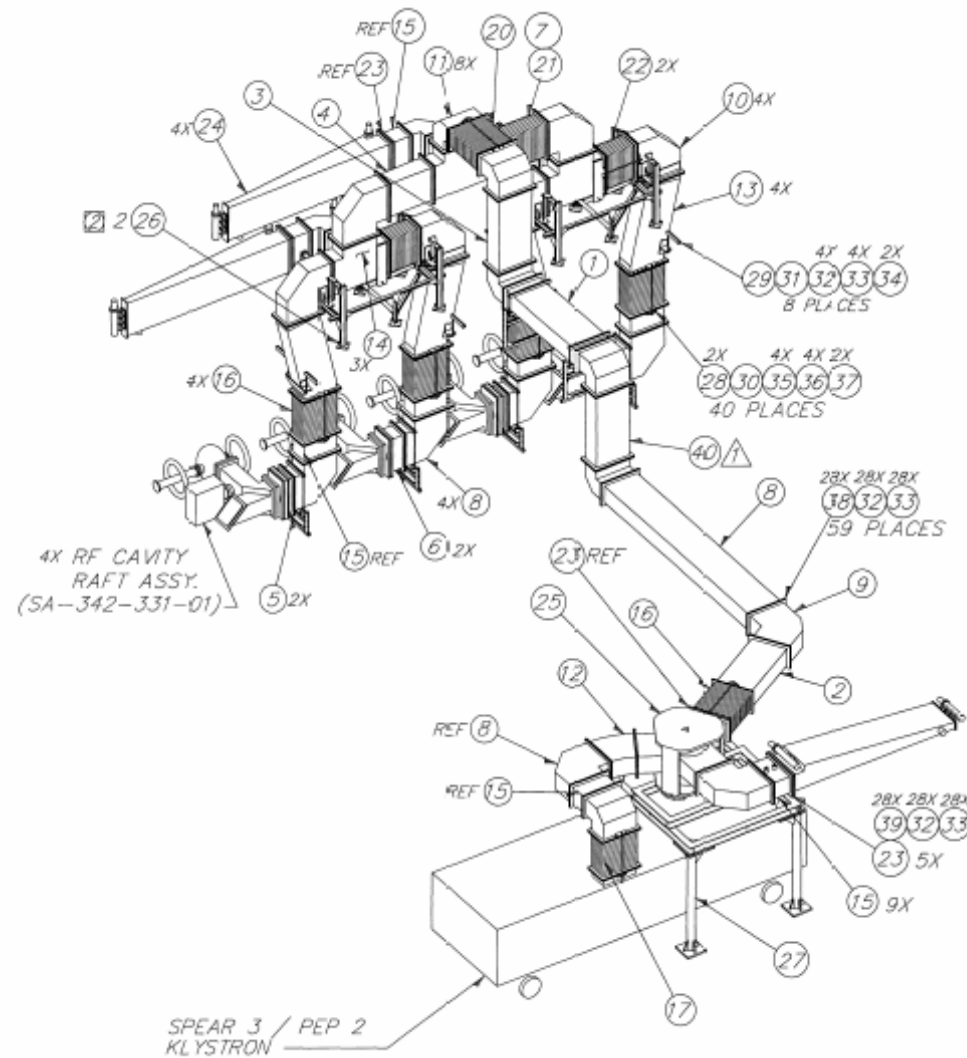
With beam energy $E_b = 3.0 \text{ GeV}$, bend radius $\rho = 7.86 \text{ m}$,
total beam power loss is $1.16 \text{ MV} \cdot 500 \text{ mA} = 510 \text{ kW}$
in 2003, and $1.33 \text{ MV} \cdot 500 \text{ mA} = 665 \text{ kW}$ in 2012
as the insertion devices are added on.

Spear 3 Beam Lifetime

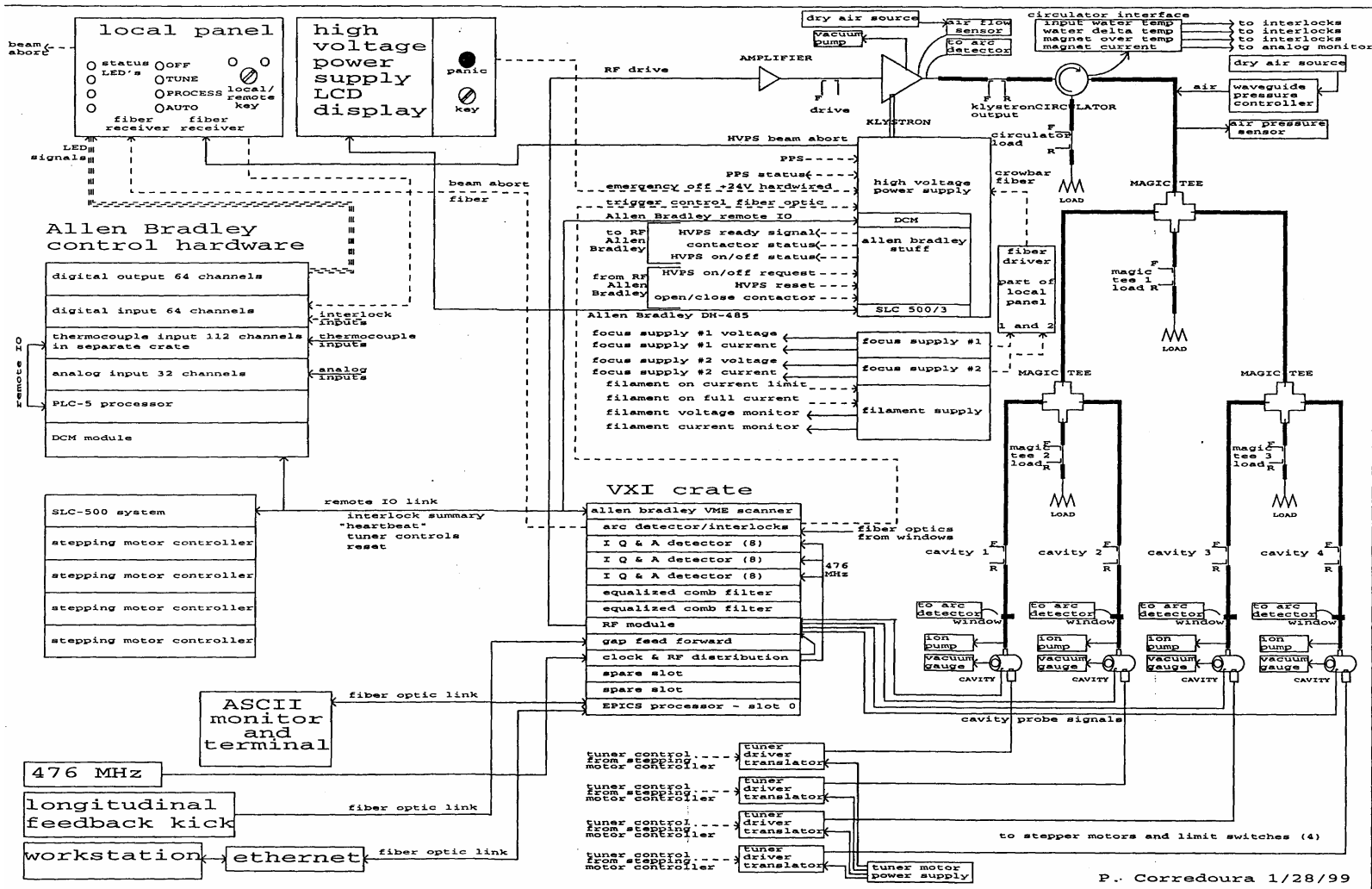




Spear 3 RF Installation



SPEAR 3 Overall System





Klystron (Repaired Marconi)

- Maximum RF Power : $P_{rf} = 1.2 \text{ MW}$
- Beam Power : $P_b = V_b * I_b = 82 \text{ kV} * 23.5 \text{ A} = 1.93 \text{ MW}$
- Microperveance $\mu P = I_b / V_b^{1.5} * 10^6 = 1.00$
- Efficiency $\eta = P_{rf} / P_b = 62\%$
- Gain $A = 10 * \text{Log}_{10} (P_{rf} / P_{drive}) = 45 \text{ dB}$
- Drive amplifier power $P_{drive} = 40 \text{ W}$
- Cathode heater power $P_h = 110 \text{ Vac} * 5.2 \text{ A} = 570 \text{ W}$
- Focusing magnet power $P_m = 70.2 \text{ V} * 47.5 \text{ A} = 3.33 \text{ kW}$
- No bucking coil power
- LCW flow for 1.5MW : 275 gpm, 150 psi, $32 \pm 1 \text{ }^\circ\text{C}$
- 2 VacIon pumps, 8 L/s each



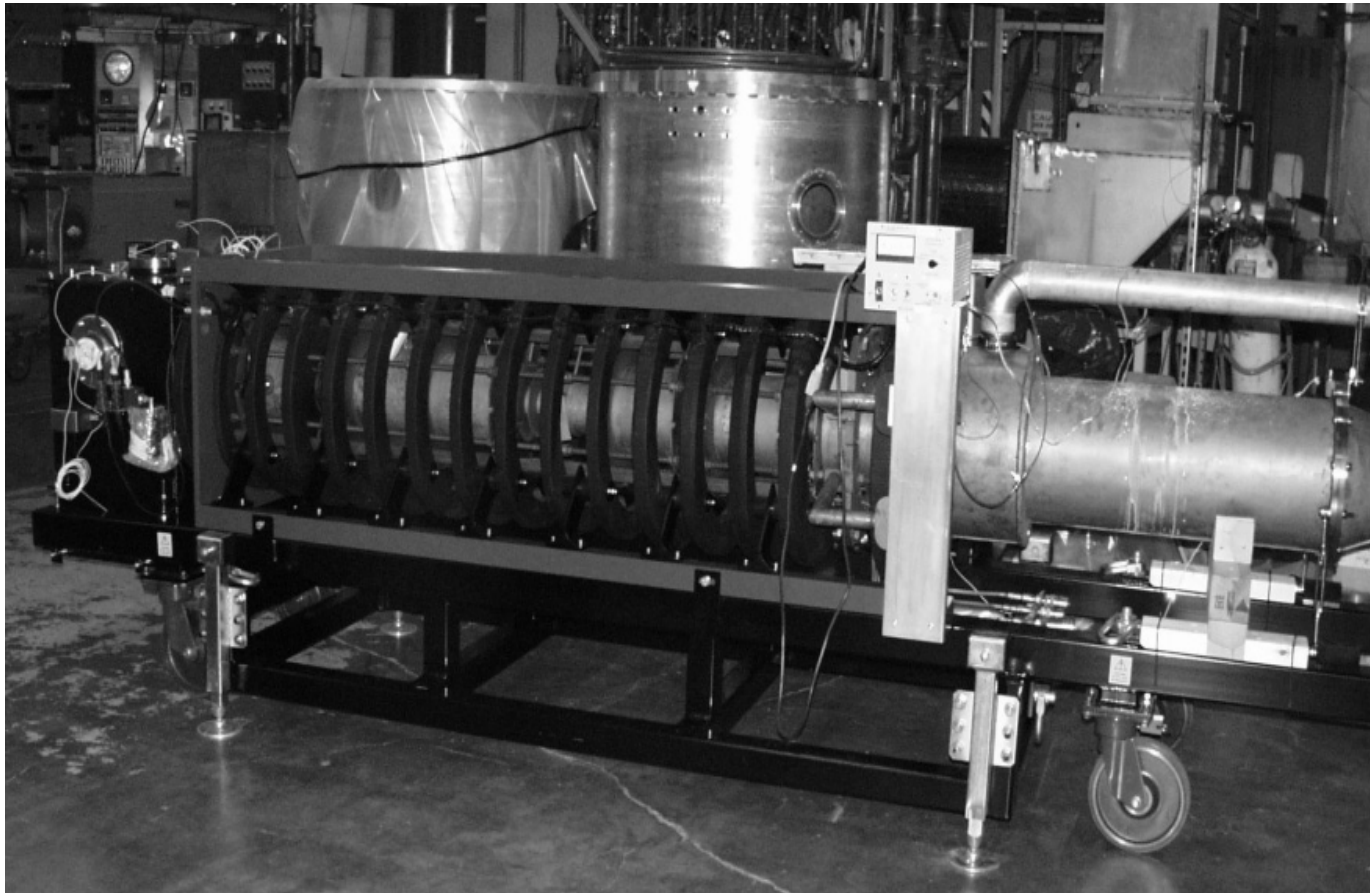
SPEAR 3 Klystron

- Spear3 klystron from Marconi
- That klystron was loaned to PEP2
- The klystron failed, and rebuilt by PCI
- SLAC Klystron Dept to produce 4 klystrons
- Those SLAC klystrons have higher power capability

Philips/EEV/Marconi Klystron Experience at SLAC

No.	Klystron	Date failed	Fil. Hrs	Failure type	Remedy
1	Philips #5	09/25/00	14,102	Heater short	Rebuilt at CPI
2	Philips #5	03/29/01	13,895	Anode dislocation	
3	Philips #5	05/22/01	5,740	Anode dislocation	Rebuilt at SLAC
4	Marconi #3	07/17/01	1,350	Vacuum leak (up to 10 mA pump current)	Rebuilt at CPI
5	Marconi #2	07/26/01	4,730	Vacuum leak (up to 60 mA pump current)	Rebuilt at CPI

Marconi Klystron



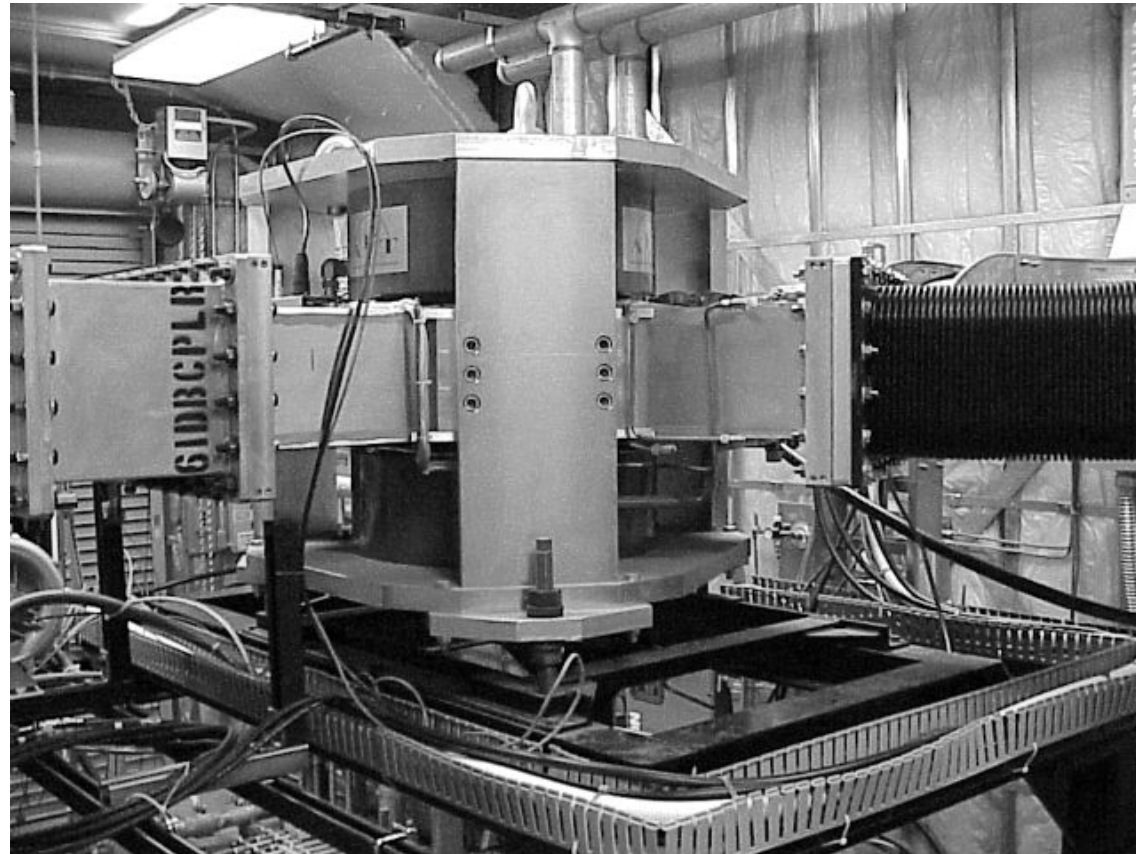


ATF Circulator Specification

- Type: Y-Junction 3-port Circulator
- Frequency : 476 ± 10 MHz
- Forward Power : 1.2 MW cw
- Reverse Power : 1.2 MW cw
- Insertion Loss : < 0.1 dB (VSWR : < 1.1 , power reflection $< 0.25\%$)
- Isolation : > 26 dB (> 14 dB in ± 10 MHz)
- Cooling LCW : > 26 gpm (150 psig, $25 \sim 40^{\circ}\text{C}$, nominal $35 \pm 1^{\circ}\text{C}$)
- Mounting Orientation : any



AFT Circulator





Water Load Specification

- Coolant : HCW (0.75% Corr-Shield by volume to LCW)
- Coolant supply : 150 psig, 10~70 °C
- Coolant return : 15 psig, <80°C
- Coolant duct : Teflon tubing
- Frequency : 476 ± 10 MHz
- Power : <1.2 MW average (<2.0 MW peak for 100 μ s)
- VSWR : <1.05 (reflected power < 0.06%)
- RF Leakage : < 0.1 mW/cm²
- Length : 9.5 feet overall
- Air pressure : <0.5 psig (0.25 psig nominal)



Water Load





HCW Station behind Booster



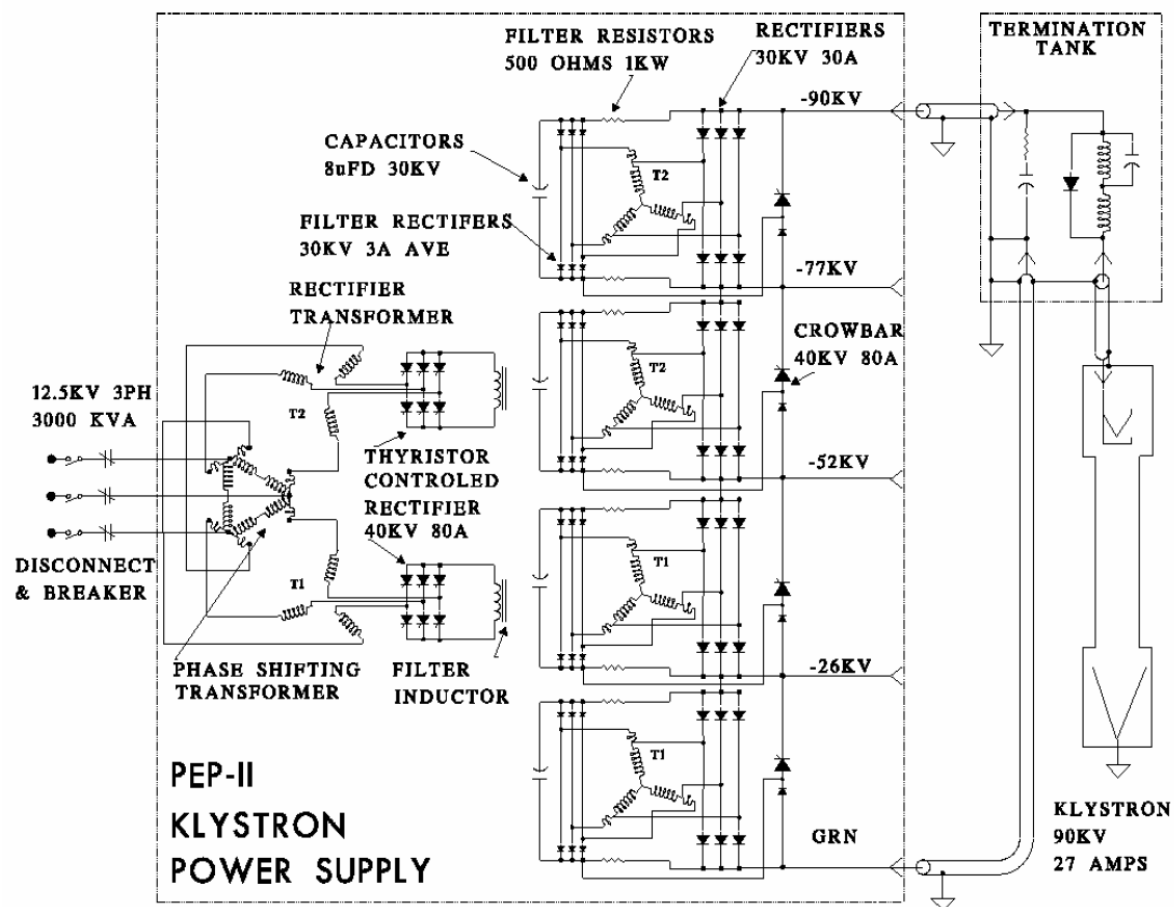


RFHV Power Supply Specification

- Output DC power : $90 \text{ kV} * 27 \text{ A} = 2.43 \text{ MW}$
Corresponds to micropervance of 1.00
and $2.43 * 0.62 = 1.50 \text{ MW}$ RF power
- Input AC power : 12.47 kV line-to-line, 127 A per phase
Power supply efficiency = $2430 / (1.73 * 127 * 12.47) = 0.89$
Lower efficiency at lower output voltage/power
- New filtering capacitors by General Atomics
- Light triggered crowbar SCR's
- Less than 0.5 Joules to the klystron in case of arcing
at 80 kV per swinging ball test of crowbar



RFHV Power Supply Schematic





Spear3 RFHV Power Supply



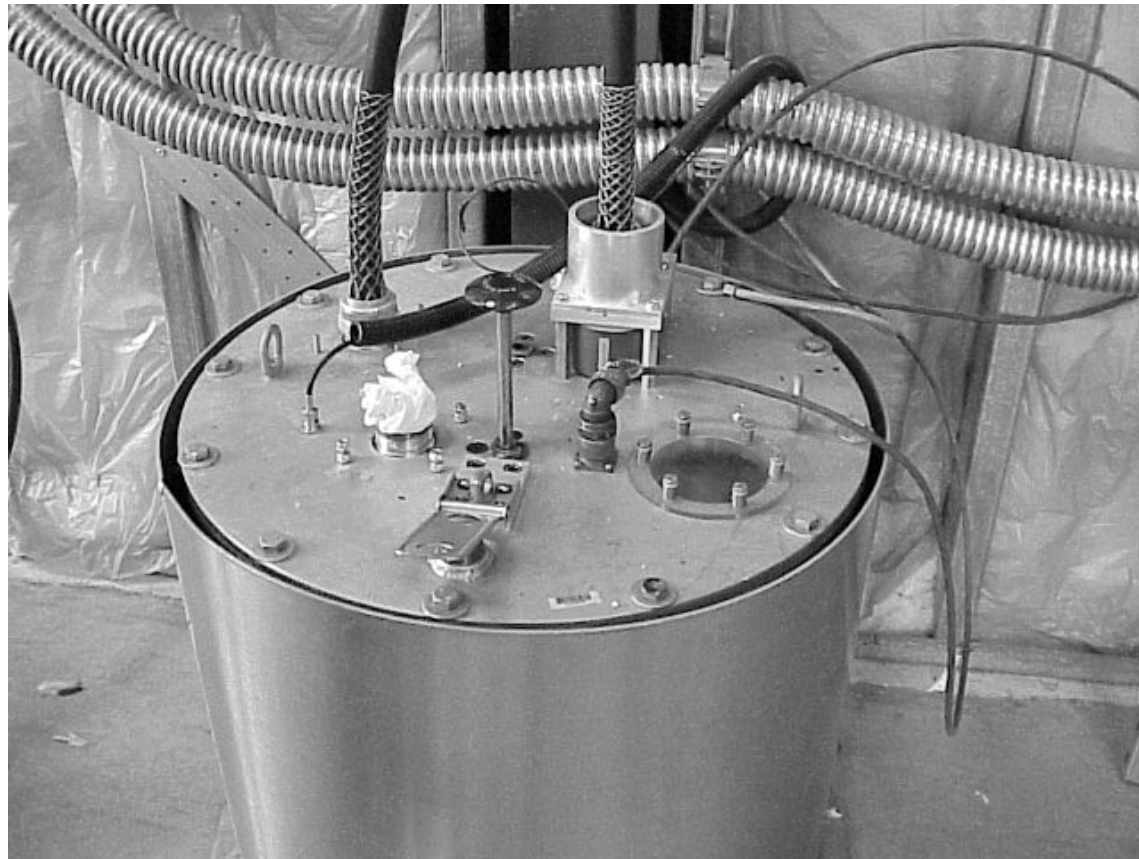
Spear3 RF System

Sam Park

11/06/2003

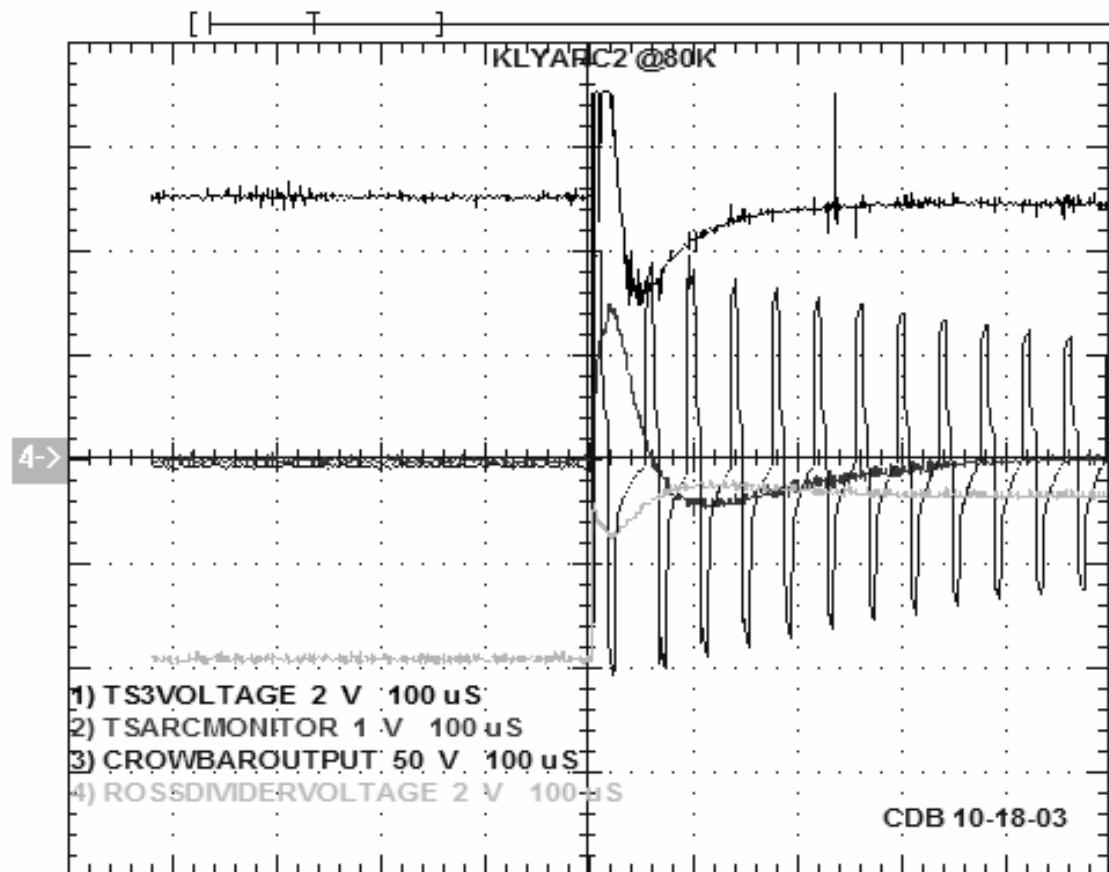


Spear3 RFHV Power Supply Grounding Tank





RFHV PS Swinging Ball Test



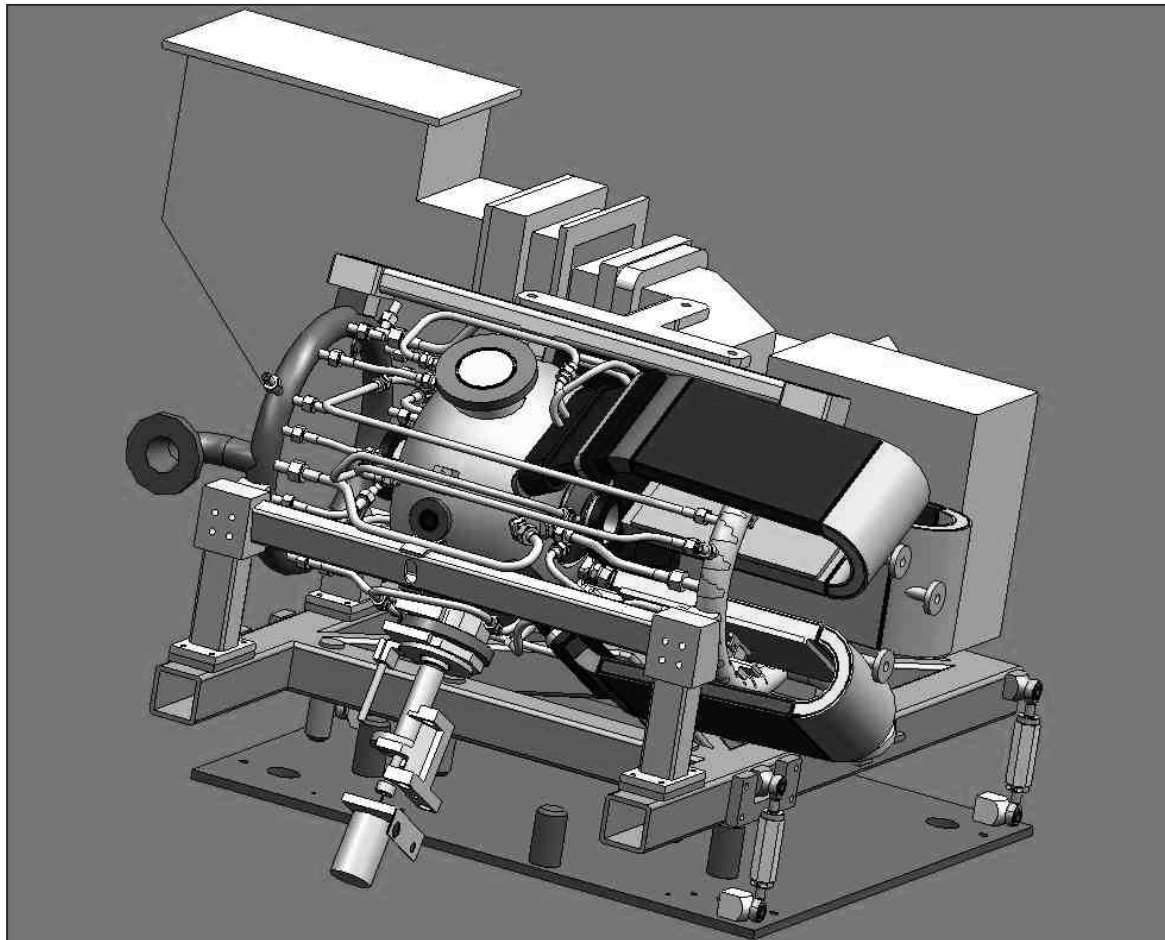


Spear3 RF Cavity Characteristics

- Frequency 476.3 MHz (different from PEP2 476.0 MHz)
- Shunt Impedance $R_a = V_g^2/P_{rf} = 7.62 \text{ M}\Omega$ (95 kW for 0.85 MV)
- Acceleration field $\sim 3.9 \text{ MV/m}$
- Coupling $\beta = 1+P_b/P_c = 3.8$ (high reflection at lower current)
- Window power $< 410 \text{ kW}$, Wall power $< 80 \text{ W/cm}^2$
- 3 high power HOM loads at each of 4 cavities
- One HOM filter per cavity at the waveguide coupler
Similar filters were used at Spear2
- One movable tuner per cavity
- Coupler window temperature is monitored by IR sensor
- $Q \sim 30,000$ at operating temperature (Fill time is $Q/\omega \sim 10 \mu\text{s}$)
- If RF is turned off on orbit interlock trip, beam is lost in $\sim 300 \mu\text{s}$

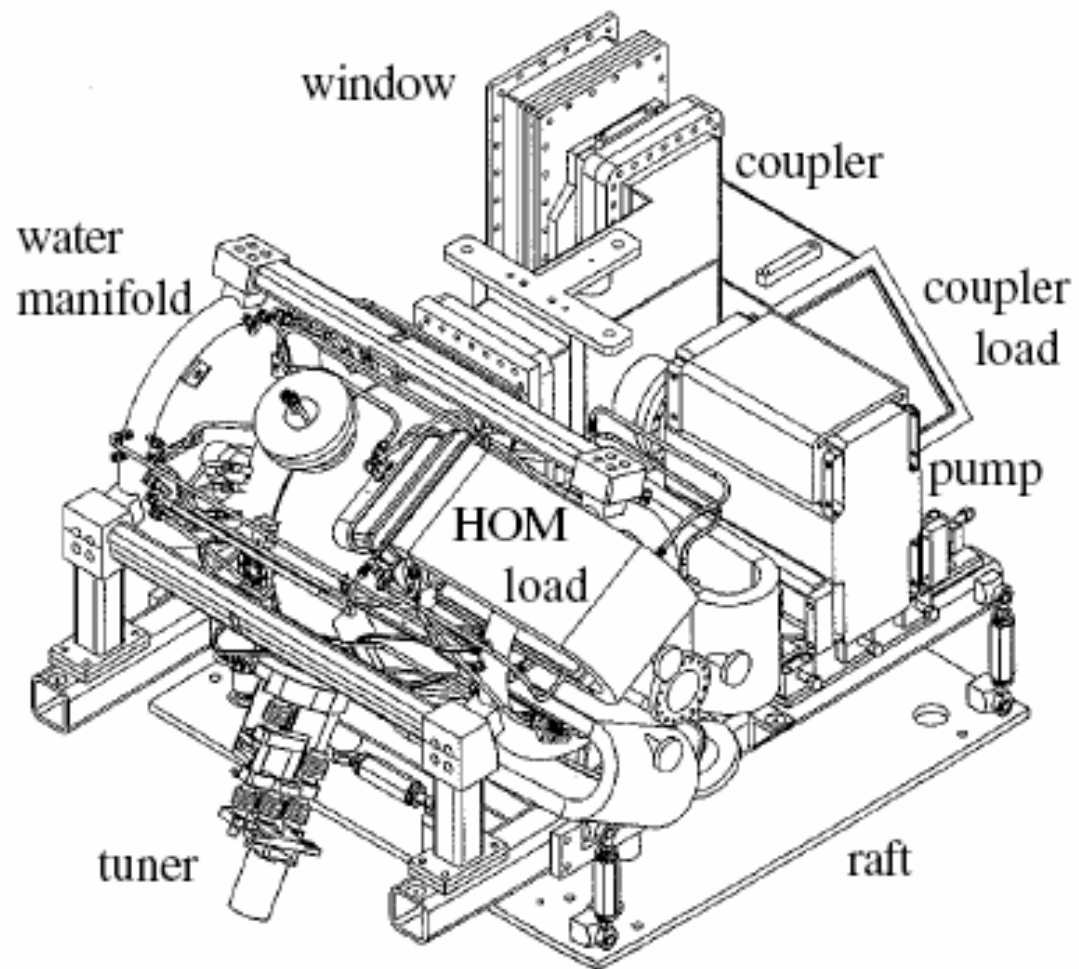


Spear3 RF Cavity Assembly



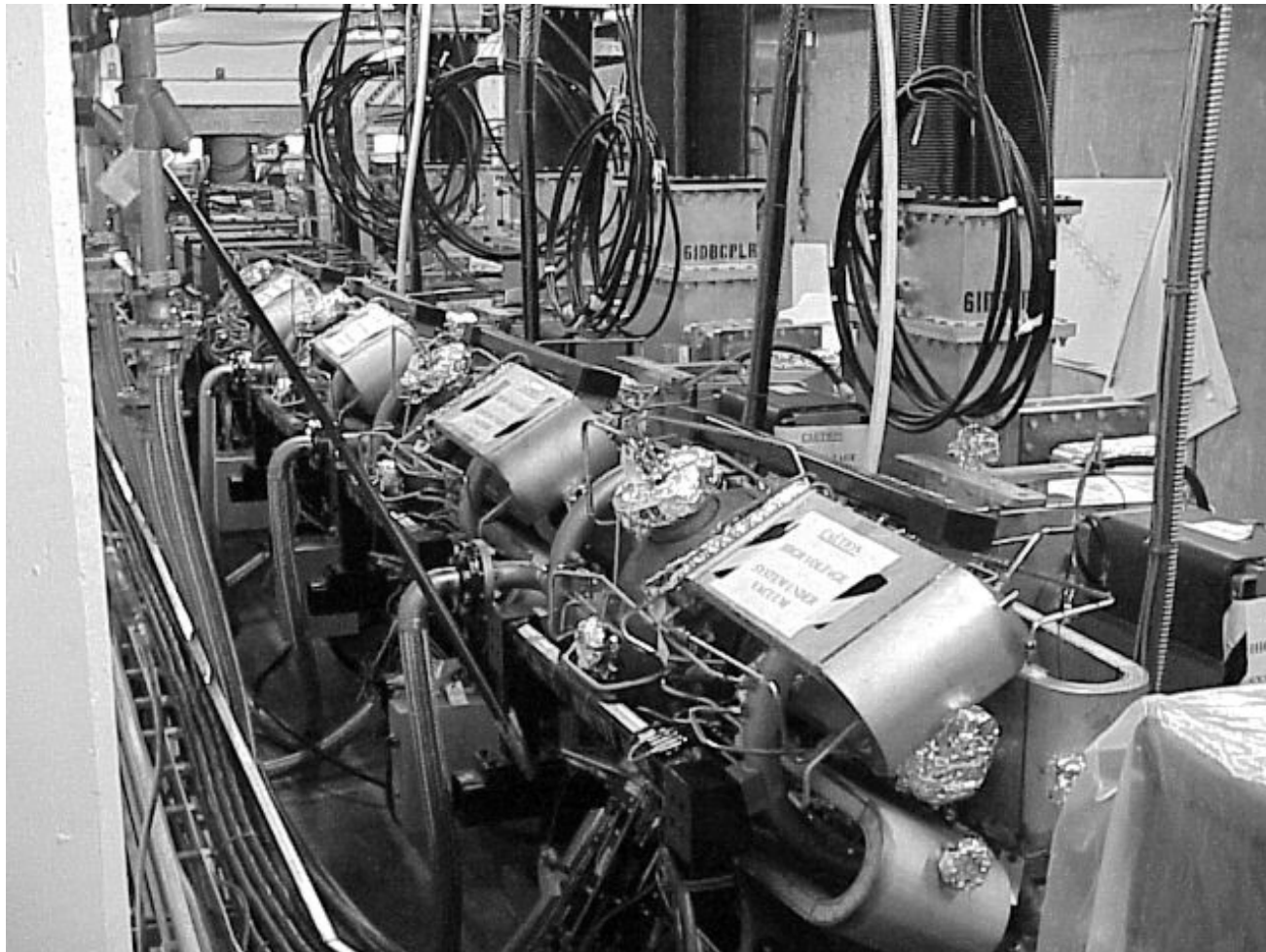


Spear3 RF Cavity Assembly





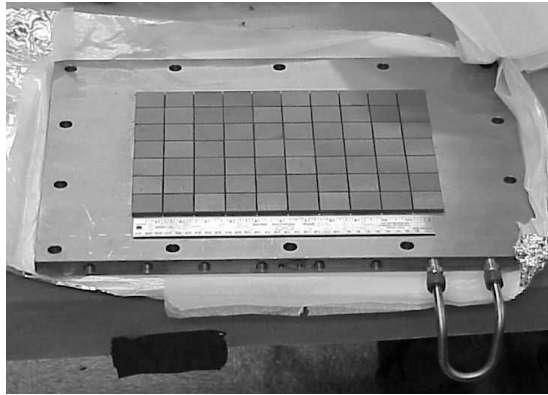
Cavities in the West Straight





Cavities in the West Straight





Matrix of 1.0 inch square ferrite tiles. They are soft- soldered onto a copper plate.

Cooling channels were drilled out from a solid copper plate.

HOM load plate, water-cooled



HOM load at E- and H-mitre

LCW flow is 6 gallons per minute. No appreciable ΔT is detected, but the flow is interlocked.

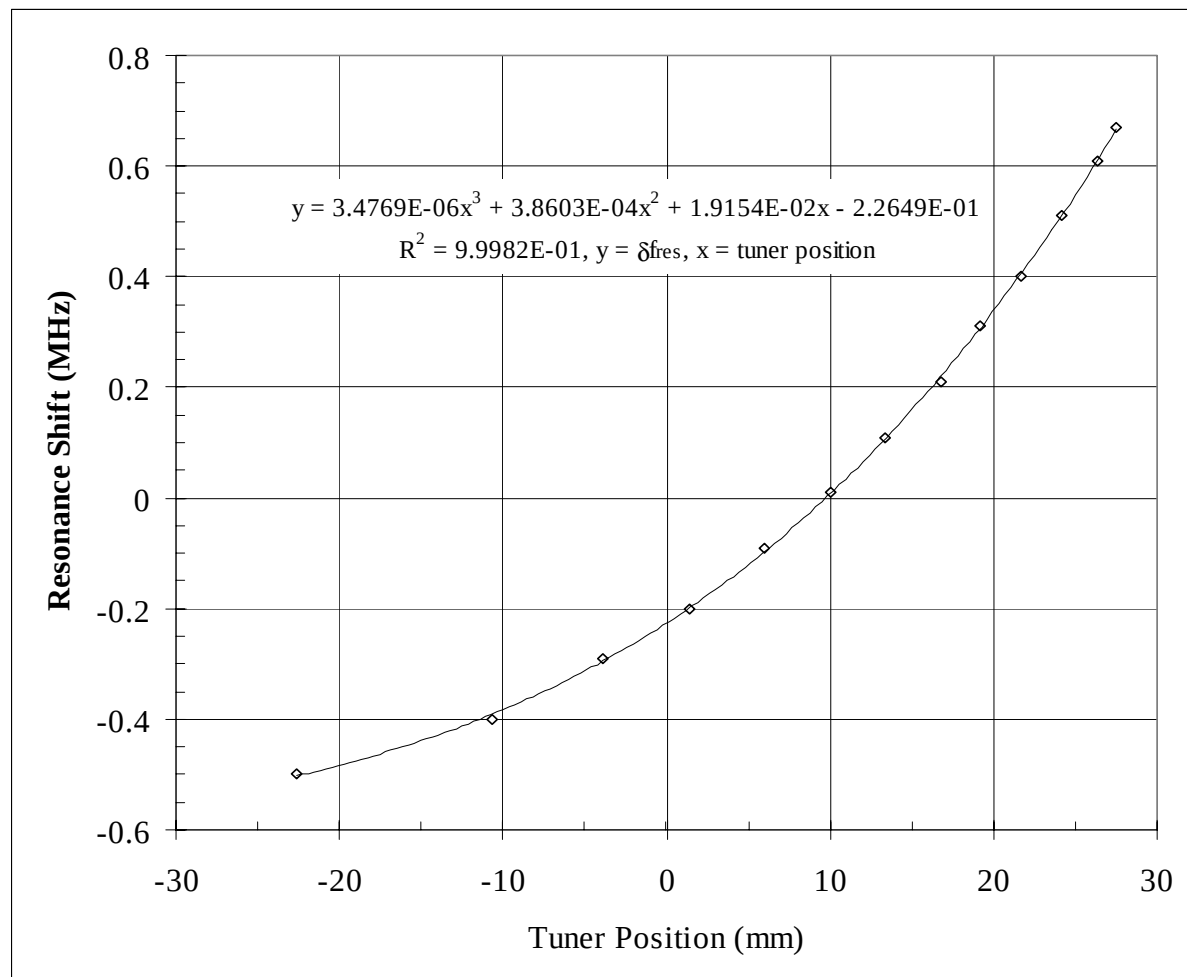


Movable Tuner below the Cavity





Movable Tuner Tuning Range



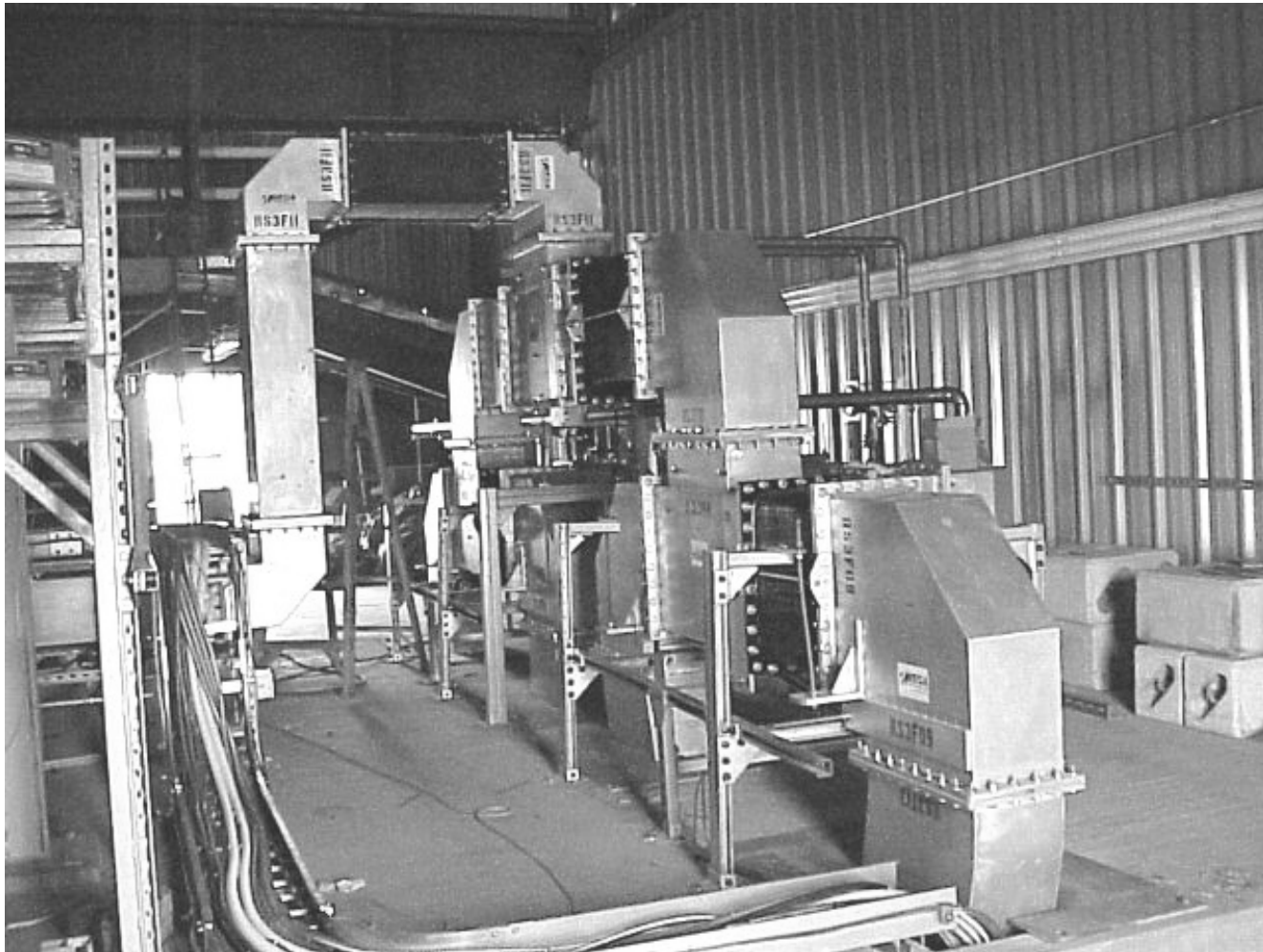


Waveguide Network & Phasing

- Magic Tees : Divide RF power evenly.
- Magic tee loads are to compensate for any mismatch and absorbs reflected power (two arms are 90 degree apart)
- Bellow lengths are adjusted to match the RF phase in cavities
- Guided wavelength $\lambda_g = \lambda_0 / [1 - (\lambda_0 / 2a)^2]^{1/2}$, $\lambda_g = c/f$
- Waveguide sections are positively pressurized with dry air to ensure that there is no mechanical gap (no RF leakage) and no moisture enters into the system
- Window at the klystron is cooled by forced air

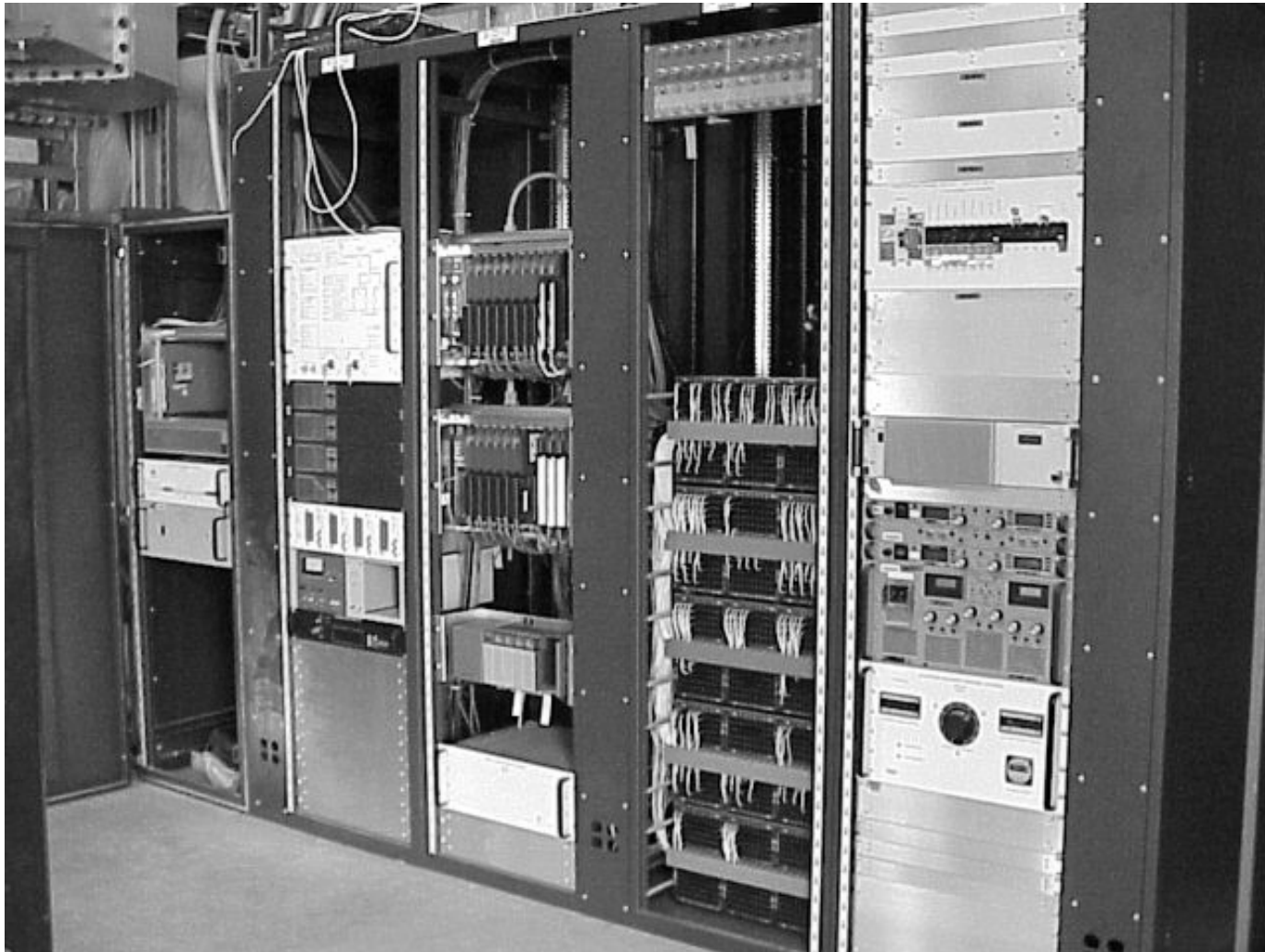


Magic-T and Bellow Network



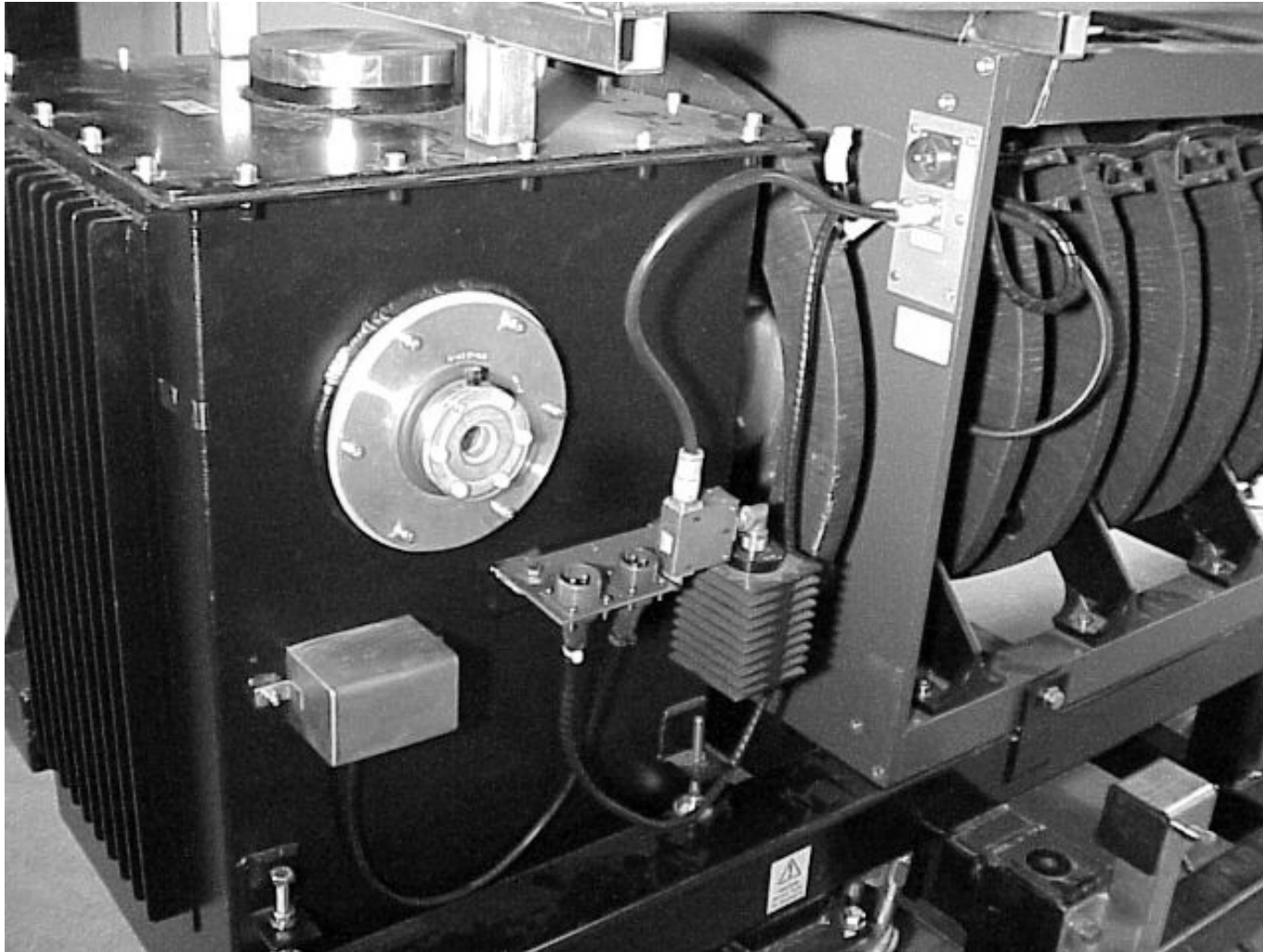


LLRF in Room 101, Bldg 132



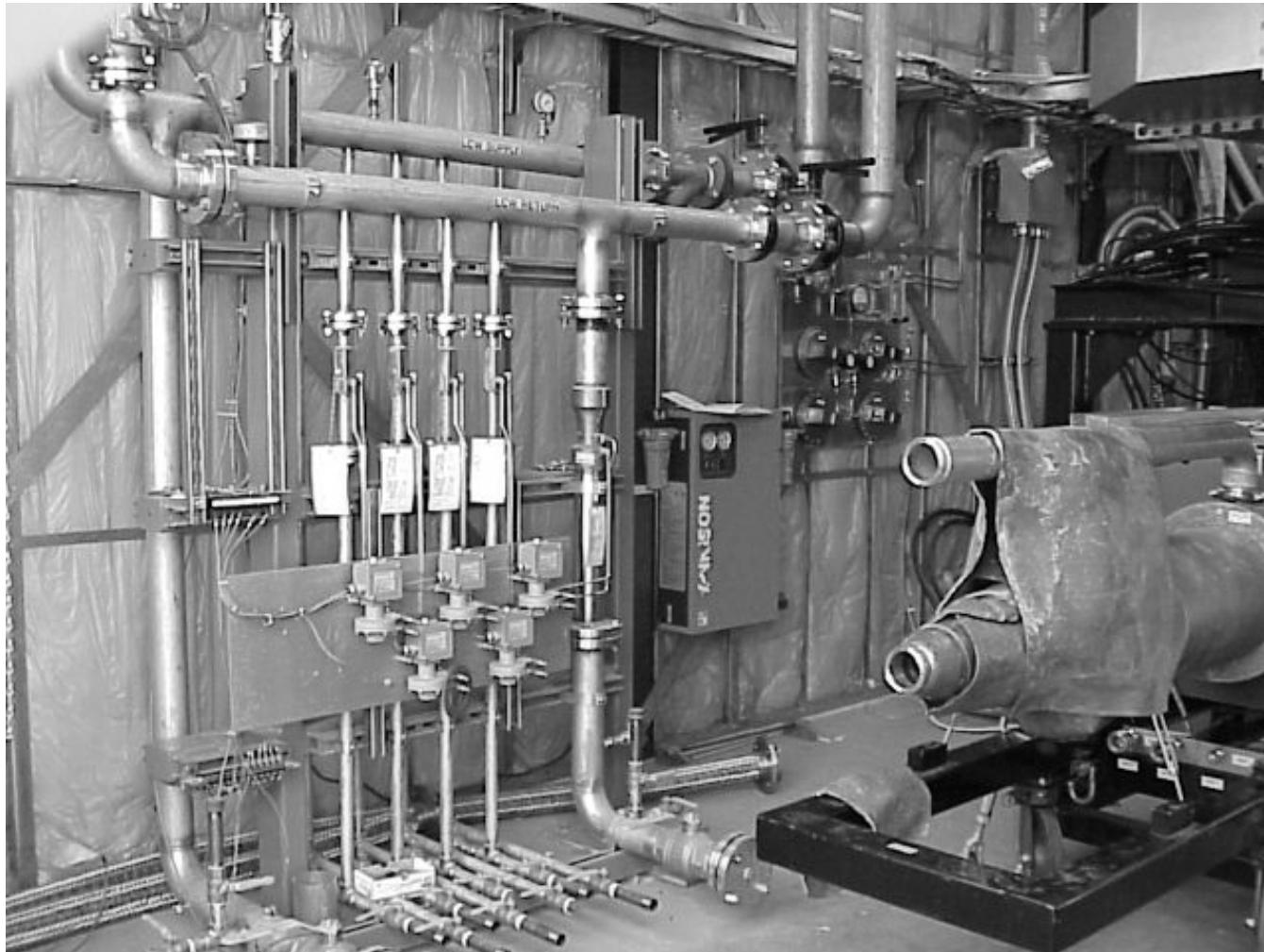


Connections to Klystron



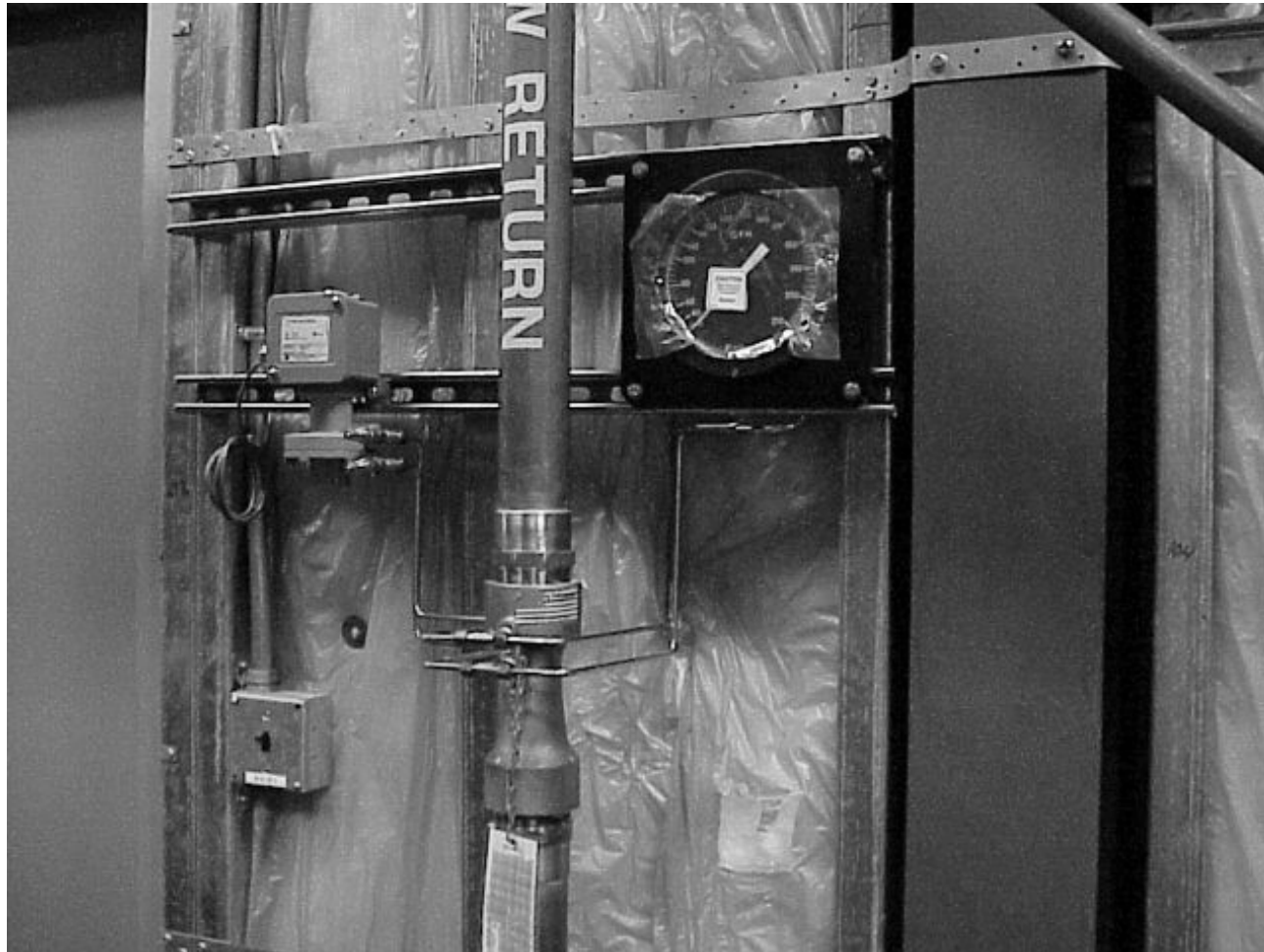


Connections to Klystron



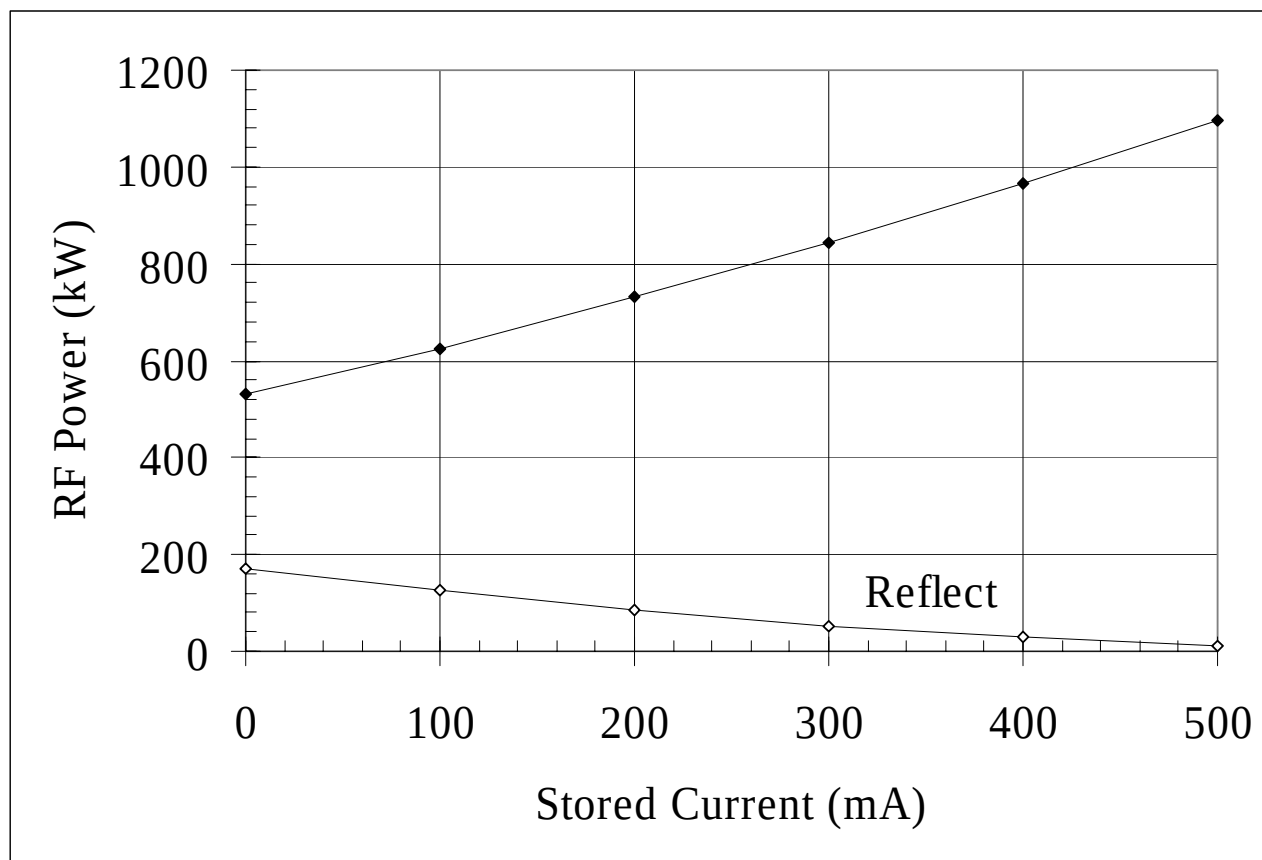


Flow monitor and interlock





Power Balance with Beam Loading





HER RF 12-3 Station		Beam Status	DB OFF		Reset	Print	Exit
		Beam Abort	DB OFF			Help	
Klystron	Circulator	HVPS	Waveguide	Cavities	MATLAB Apps		
Vacuum	Flow	Temps	Phase/Pwrs	Arc Intlks	MeasBmPhs		
System	Tuners	Feedback	Modu Diags	Stn Diags	MoreMATLAB		
In ON_CW. Direct/Comb loops on.							
State Control:	Auto Reset Tries Left	20	Local Pnl Keyswitch	DB ONLINE	Contactor	DB ON	Close
OFF	Max Cav Vacuum (Torr)	5.0e-09	Stn Gap Volt (kV)	2800	Stn Phase (Deg)		
PARK	Max Cav Gap Volt (kV)	700	TUNE Drive Pwr (W)	55.0	-200.0 -78.9 200.0		
TUNE	Max Klys Fwd Pwr (kW)	450	ON_CW Drive Pwr (W)	30.0	Swept Sine: Beam Tickle:		
ON_FM	HVPS Loop	OFF PROC ON			100 HZ 1000 HZ OFF		
ON_CW							
Filament Voltage (V)	DB 78.8	Klystron		HVPS Voltage Control (kV)		HVPS Voltage (kV) DB 67.32	
Filament Current (Amp)	DB 8.00			0.00 67.29 90.00		HVPS Current (Amp) DB 18.27	
Filament Time Left (Sec)	DB 0			Max Voltage (kV) 72.00		HVPS Power (kW) DB 1230	
Drive Power (W)	DB 30.5					Collectr Power (kW) DB 852	
Vacuum Level (Torr)							
Forward Power (kW)	DB 378						
Reflected Power (kW)	DB 1.7						
Stn Gap Volt (kV)	DB 2801						
Cavity A Volt (kV)	DB 708						
Cavity B Volt (kV)	DB 694						
Cavity C Volt (kV)	DB 706						
Cavity D Volt (kV)	DB 694						
		Tuner Posn (mm)		-30.00 20.00			
		A DB -9.74					
		B DB -6.41					
		C DB -0.50					
		D DB 2.83					

HER RF 8-3 Klys and Circ

Reset

Print

Exit

Go To

Help

Station

HVPS

Waveguide

Cavities

Tuners

Feedback

Type of Klystron

DB PHILLIPS

Max Voltage (kV)

DB 83.30

Vacuum
uAmp

DB -0.00

Status

DB OK

Klys Arc

DB OK

Circ Arc

DB OK

Water Supply

Sply Temp (degC) DB 36

Delta Temp (degC)

DB 16

Rtn Flow Switch

DB OK

Collector

DB 0

DB OK

Klys Body 1

DB 2

DB OK

Klys Body 2

N/A for Phillips

DB 2

DB OK

Circulator

DB OK

DB OK

Circ Input Temp

DB OK

Circ HCW Supply

Sply Temp (degC) DB 21

Rtn Flow Switch

Load Rtn (degC) DB 20

DB OK

Klystron Window Air Supply

Delta Temp(degC) DB -5

Rtn Flow Switch

Rtn Temp (degC) DB 47

DB OK

Circ Magnet Current (Amp)

DB 1.92

Klystron Drive Power (W)

Drive Power DB 16.1

Drive Pwr Plot

Setpoint DB 14.0

Amplifier Temp (degC)

High Pwr DB 16.0

DB 24

Saturation DB 40.0

Klystron Solenoids

Main

Bucking

Voltage (V) DB 129.7

DB 3.61

Current(Amp) DB 39.6

DB 5.04

Interlock

DB OK

DB OK

On/Off Stat

DB ON

DB ON

Klystron Filament

On/Off Stat DB ON

Secs Left DB 0

Voltage (V) DB 70.2

Current(Amp) DB 7.16

Power (W)

DB 500.5

Timer Override

Override

Filament & Solenoid
On/Off

ON

HVPS

Voltage (kV) DB 73.73

Current(Amp) DB 21.90

Klys Perveance

DB 1.10

Power (kW)

DB 1615

HVPS Volt Plot

RF Power (kW)

KlysOut Frwd DB 821

KlysOut Refl DB 2.6

Klys Collectr DB 795

Circ Ld Frwd DB 1.4

Circ Ld Refl DB 0.0

Klys Eff (%) DB 51

Klys Gain(dB) DB 47.1

Klys Pwr Plot

Normal (No Beam)

Klys Power (kW)

DB 0.0

Hi Drive Setting

Klys Power (kW)

DB 700.0

Setpoint

DB 46.2000



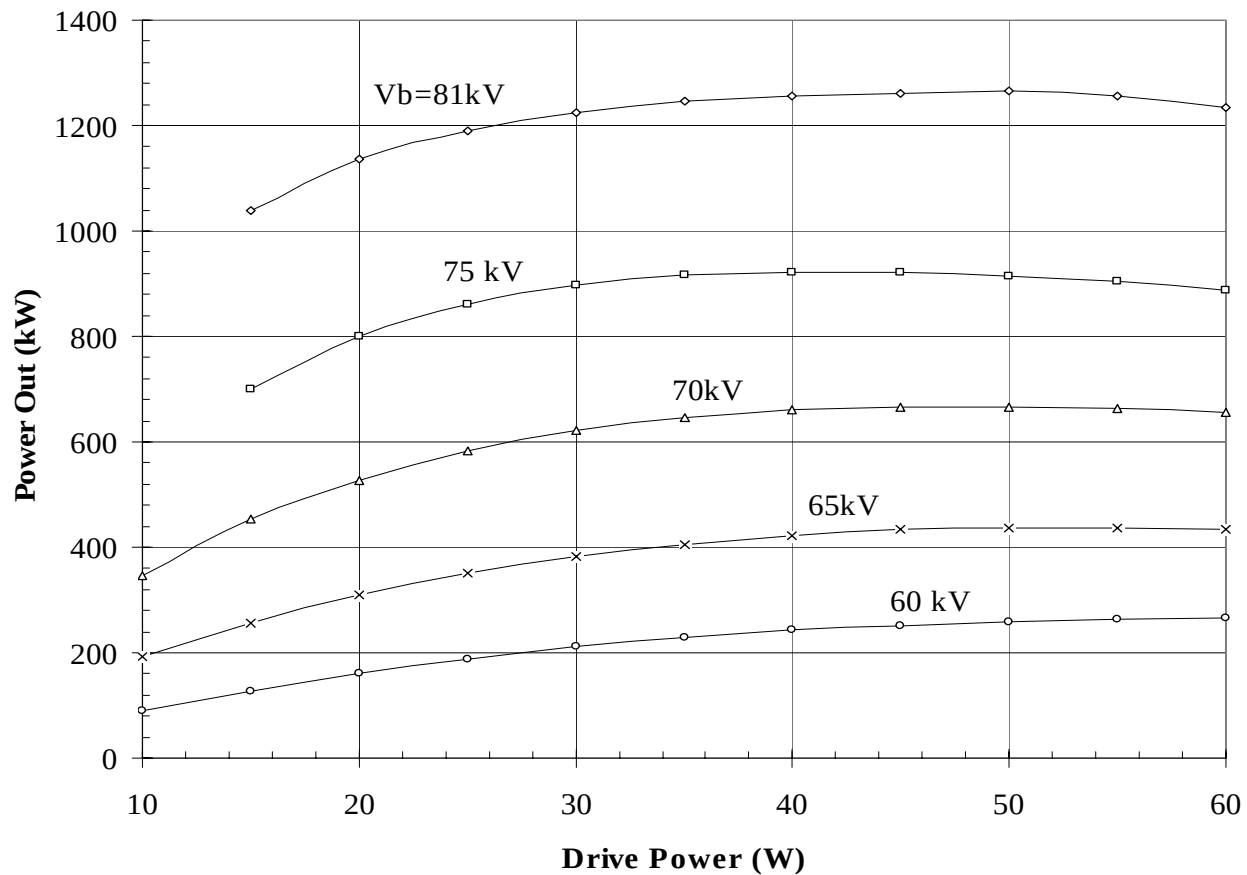
HER RF 8-3 Klystron HVPS		Reset	Print	Exit				
			Go To	Help				
Contactor	Open	Close						
	Contactor Closed		DB	CLOSED				
Contactor Open		DB	CLOSED					
Contactor Status		DB	ON					
Contactor Ready		DB	READY					
Voltage Control (kV)		HVPS Ctrl	Klystron					
DB 0.00 67.52 90.00		HVPS Volt Plot						
Stn Turn-On Voltage	DB 55	Max Volt Active S	DB	ACTIVE				
Maximum Voltage (	DB 75.00	Regulator Voltage	DB	67.52				
		Max Volt Rbck (kV	DB	75.00				
Klys Output Power					DB	397		
Klys Collectr Power					DB	836		
Klys Efficiency (					DB	32		
RF PLC Voltage (kV					DB	67.4		
RF PLC Current (Am					DB	18.50		
RF PLC Coil Power					DB	905		
Klys Perveance					DB	1.04		
HVPS Current (Amp					DB	18.27		
HVPS Voltage (kV)					DB	67.50		
HVPS Power (kW)					DB	1233		
Latching Faults								
Over Voltage	DB	OK	Over Temperature	DB	OK	Manual Reset		
Arc	DB	OK	Oil Temperature	DB	OK			
Crowbar	DB	OK	Oil Level	DB	OK			
Emergency Off	DB	OK	Transformer Press	DB	OK			
Waveguide Air Pre	DB	OK	Transformer Vac/P	DB	OK			
AC Current	DB	OK						
Supply Status					PPS			
AC Current (Amp)	DB	88.22	Enerpro Fast Inhi	DB	ON	DB	PERMIT	
12 kV Available	DB	AVAIL	Enerpro Slow Star	DB	ACTIVE	SCR		
AC Auxiliary Powe	DB	ON	Supply Status	DB	ON	SCR 1 Stat	DB	ON
DC Auxiliary Powe	DB	ON	Supply Ready	DB	READY	SCR 2 Stat	DB	ON



HER RF 8-3 Feedback		Print Exit	
		Go To Help	
Station Direct Lp Ripple Lp Drive Pwr Gap Volt Modu Diags			
In ON_CW. Direct/Comb loops on.		Station State DB ON_CW	
Loop Control	ON	ON	Tuners
Loop Gain (dB)	13.2000	14.8000	HVPS Loop
Gain Offset (dB)	0.0000	0.0000	OFF PROC ON
Conv Factor	183.0000	147.0053	HVPS Loop Status
DAC Counts	DB 836	DB 808	DB GOOD
Loop Phase (Deg)	-12.7	2.5	TUNE - DAC Control - ON CW
Phase Offset (Deg)	DB 14.8	DB -24.7	ON ON
Total Phase (Deg)	DB 2.0	DB -22.2	DAC Control Status
For Ramping ON:	1.0	5.0	DB GOOD
Gain Delta (dB)	1.0	5.0	Direct Feedback Loop Options
Init Gain Offset (dB)	-10	-15	Freq Offs Tracking Integral Compensation Lead Compensation
			ON ON ON
			Ripple Loop Gain Tracking Woofer State Kick
			ON ON Taxi Link OK
			Gap FF Loop LFB Woofer Single TAXI sync
			ON ON Send
Stn Gap Voltage (kV)	DB 2199	Stn Volt Plot	Station Phase (Deg)
Gap Volt Setpoint (kV)	2200	HVPS Volt Plot	offset 104.81 Ramp Delta 0.1
HVPS Voltage (kV)	DB 72.42	Klys Pwr Plot	Base -133.01 Monitor Delta 0.1
Klys Out Fwd Pwr (kW)	DB 749	Drive Pwr Plot	Total DB -13.4 Cav Freq Offset DB 14.8
Drive Power (M)	DB 16.3	Klys Gain (dB) DB 46.6	MATLAB Applications
Drive Pwr Setpoint (M)	14.0	Gain Setpoint (dB) 46.2000	Meas Rippl MeasDirCls
High Pwr Drive Pwr (M)	16.0		Plot Total MeasBmPhs
Saturatd Drive Pwr (M)	40.0		MoreMATLAB

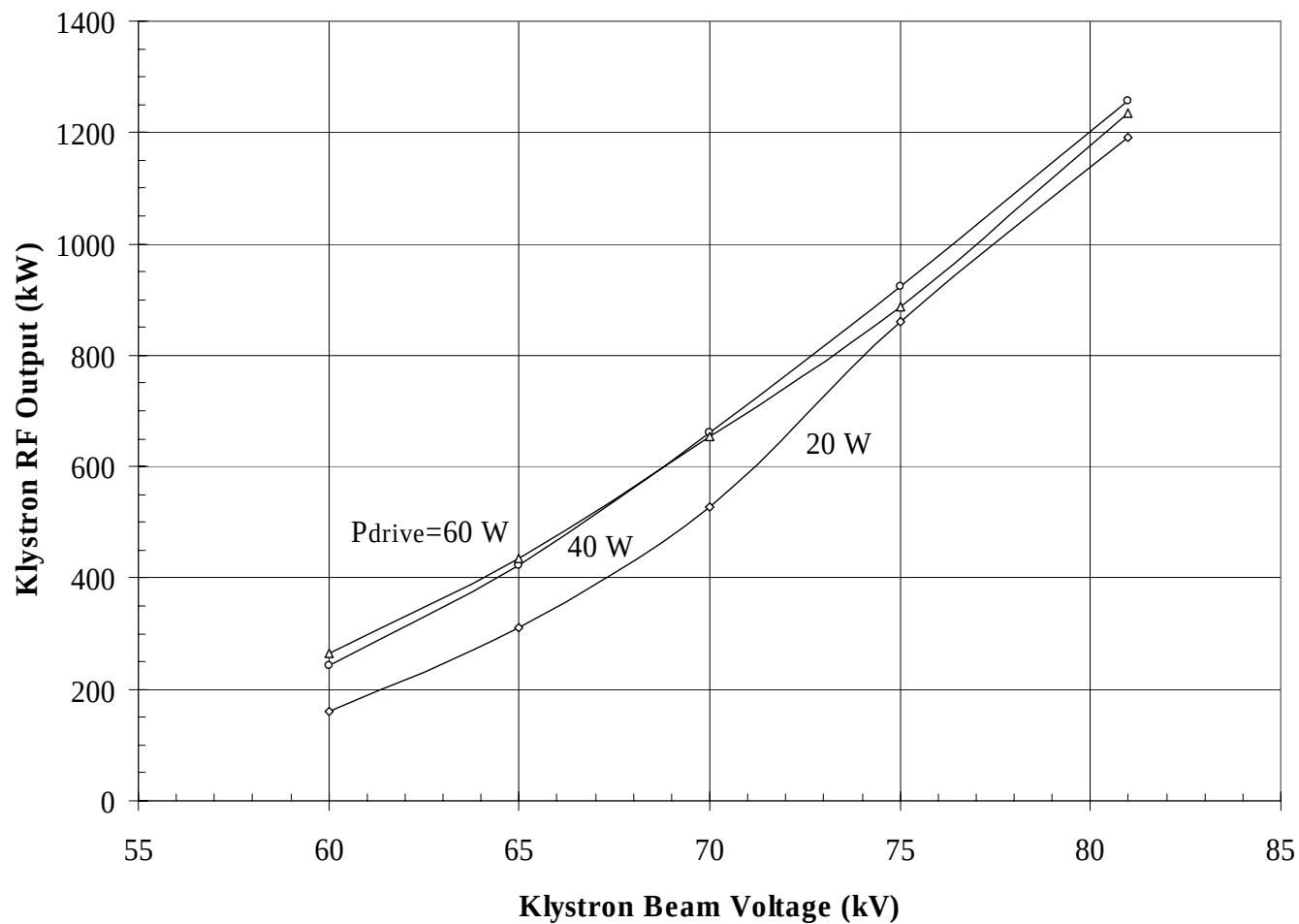


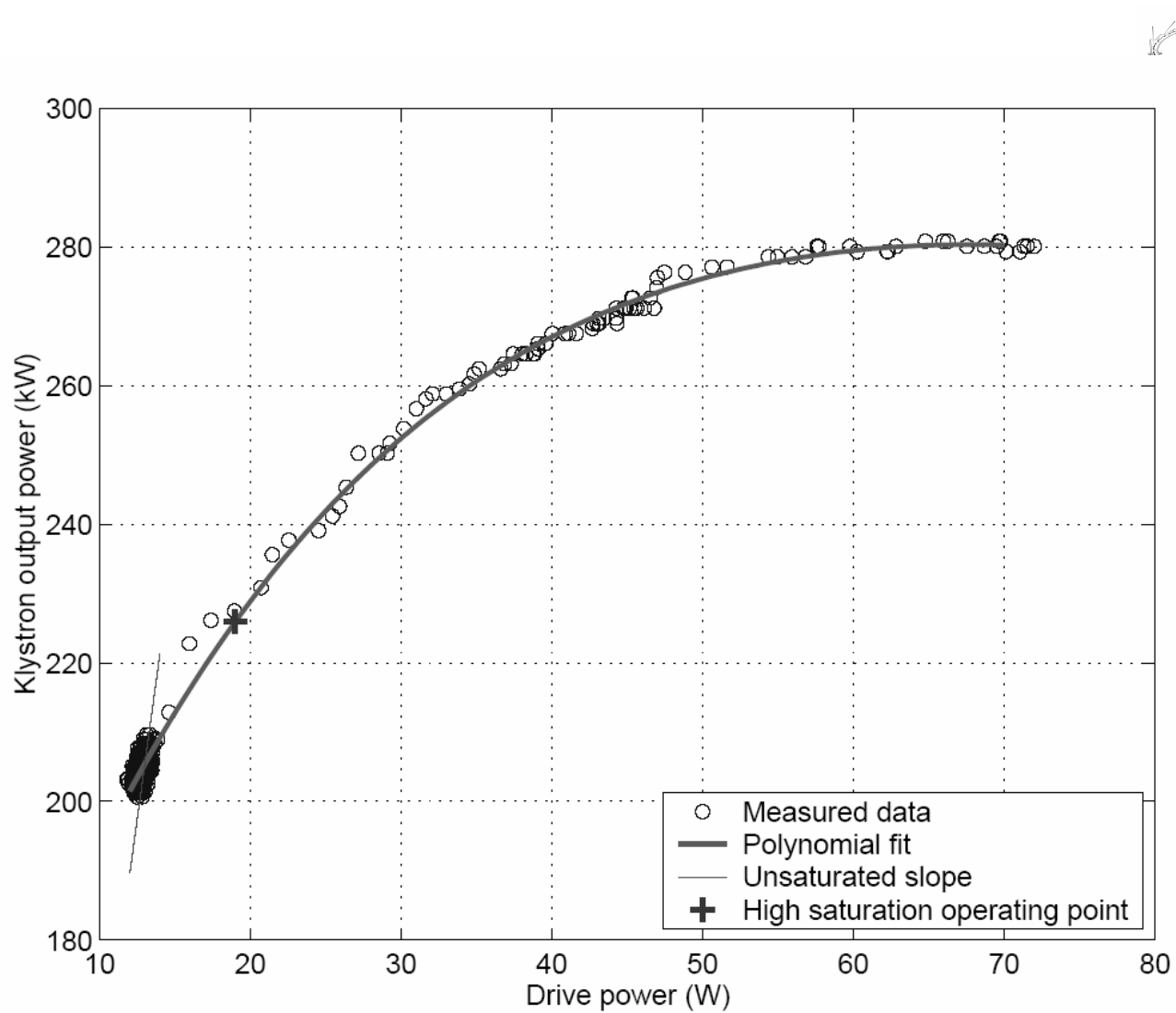
Marconi Klystron Gain Curves



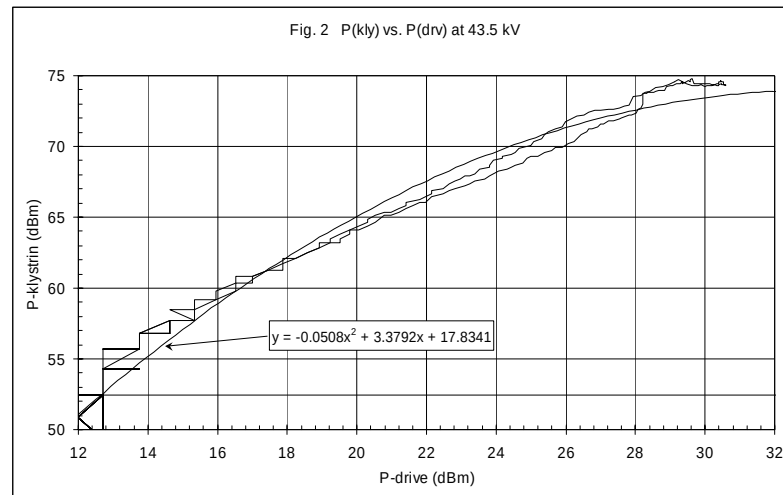
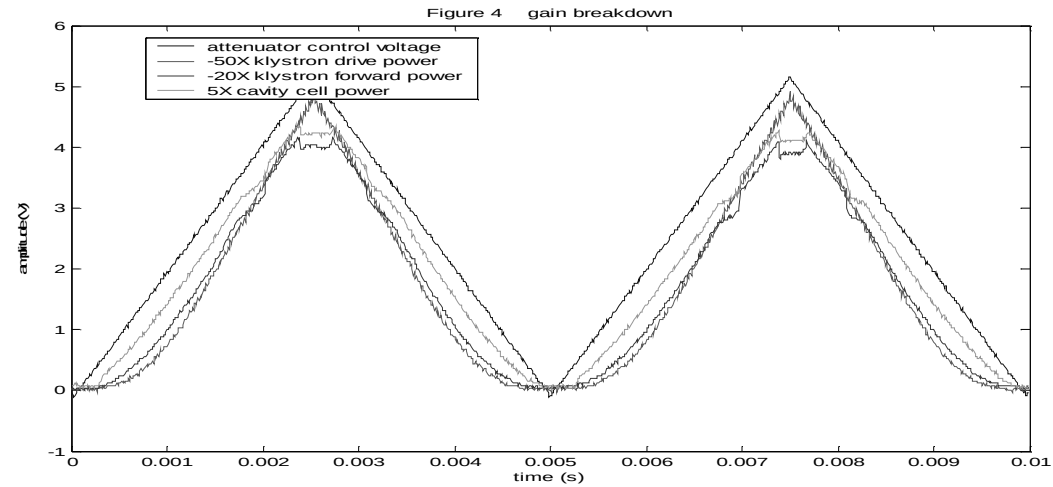


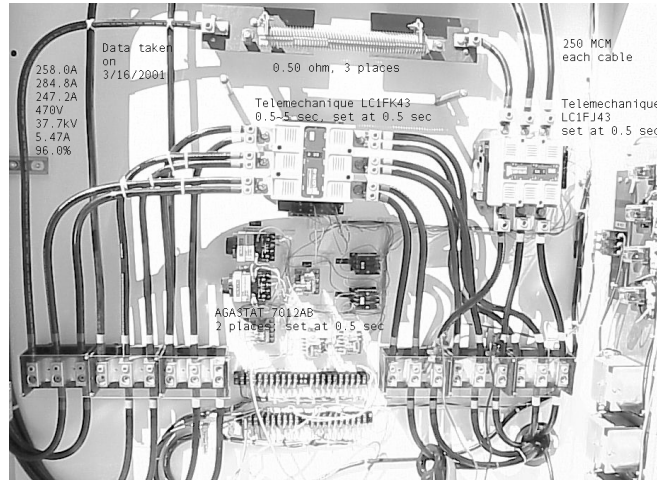
Marconi Klystron Gain Curves



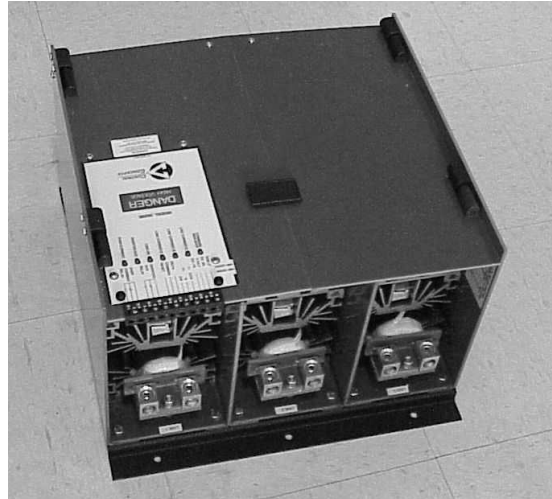


Booster klystron saturation

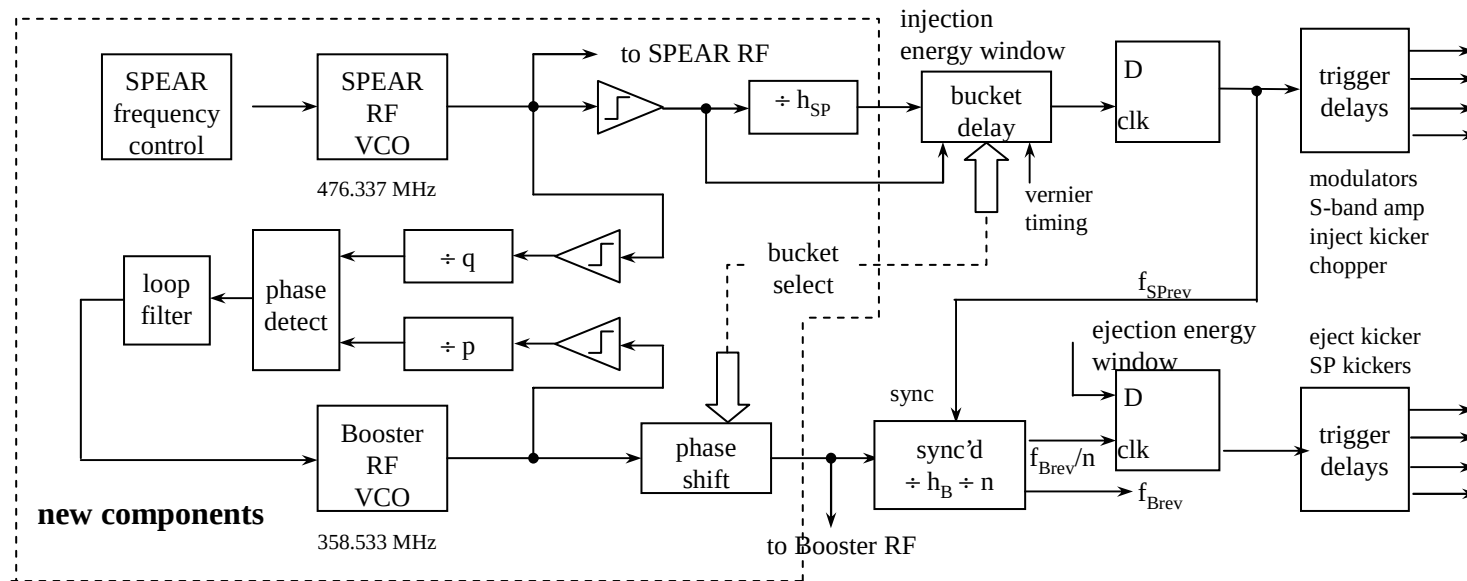




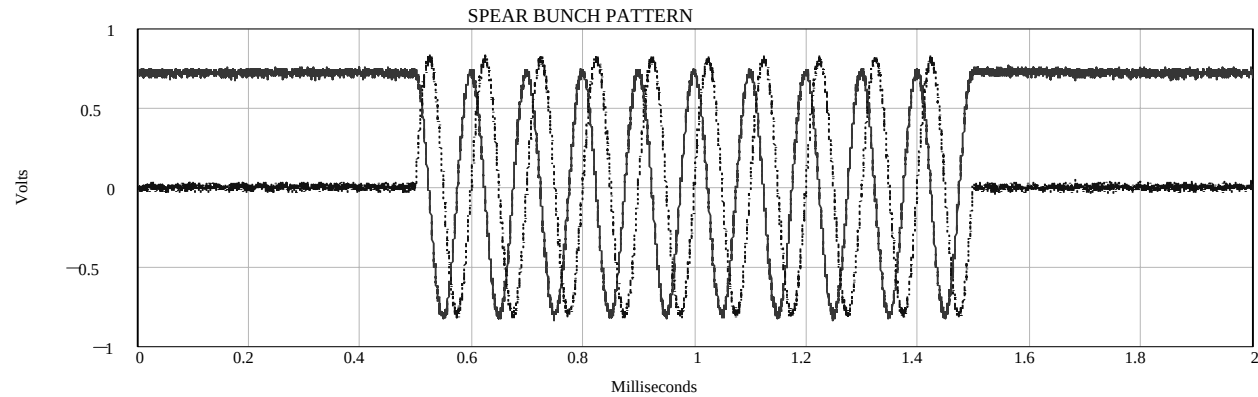
Existing Booster RF Soft-Start, Mechanical



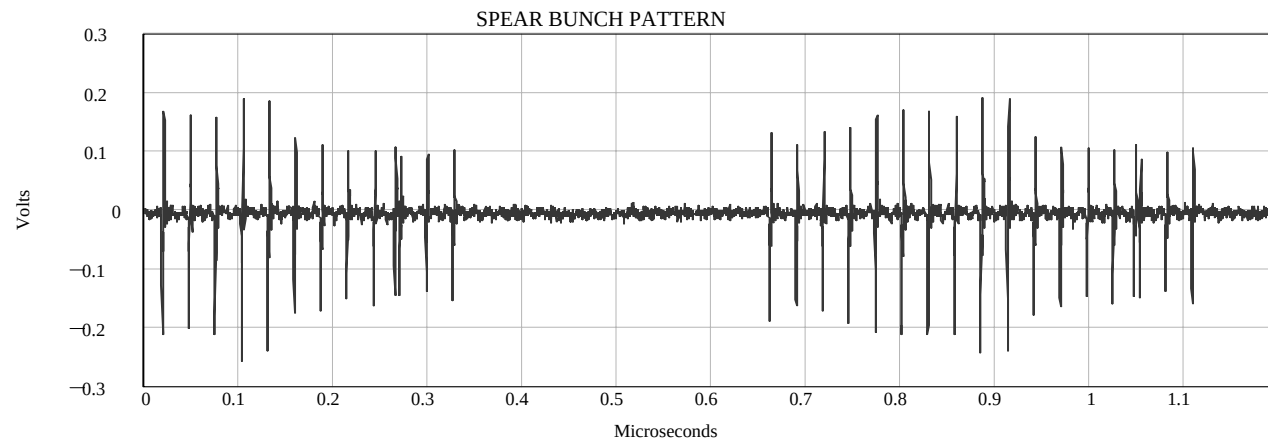
SCR Assembly with Built-In Soft Start



- Single-bunch filling
- Phase-lock Booster RF (358.505 MHz)
to SPEAR RF (476.300 MHz)
- $C_{\text{Boo}}/C_{\text{SPEAR}} = 4 / 7$
- $f_{\text{Boo}}/f_{\text{SPEAR}} = 70 / 93$



Driving I&Q Modulator



Test Fill Pattern in Spear2



SPEAR 3 Cavity Production

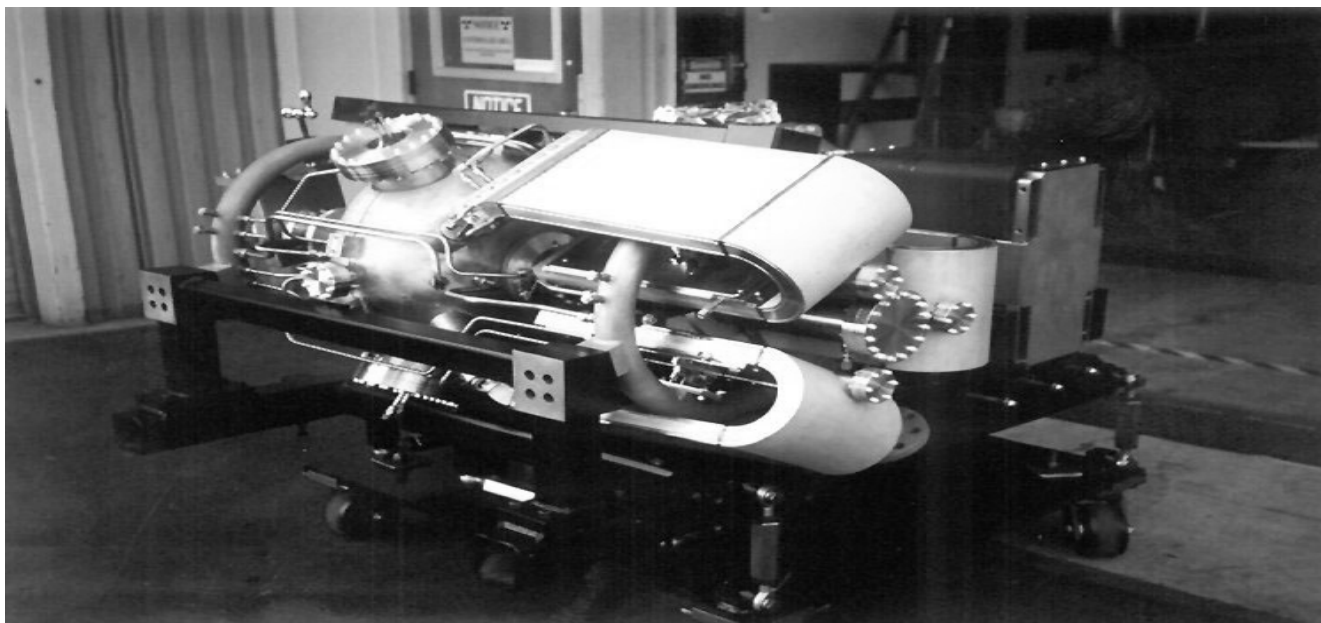


Cavity body milling at Accel



Electroforming at Accel

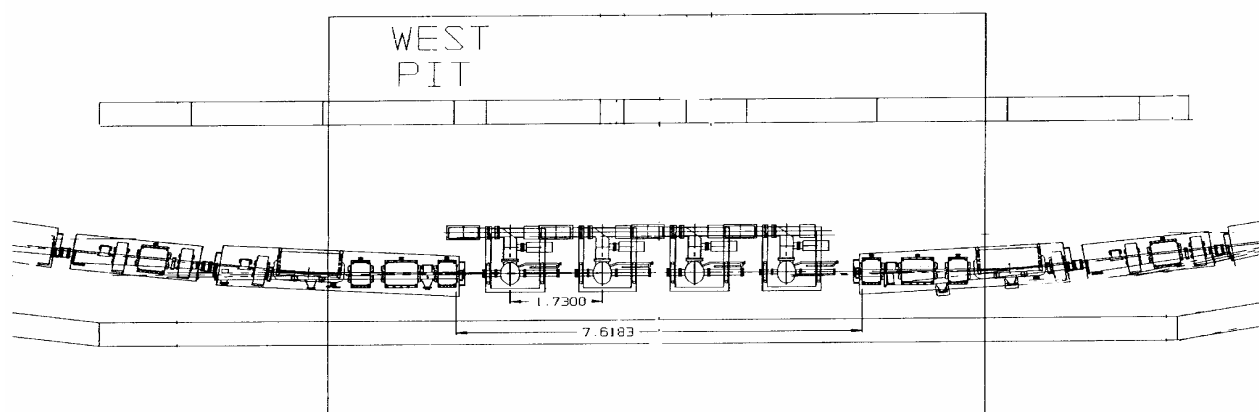
SPEAR 3 (PEP-II) RF Cavities



BR_040

RF Cavity

8-19-97

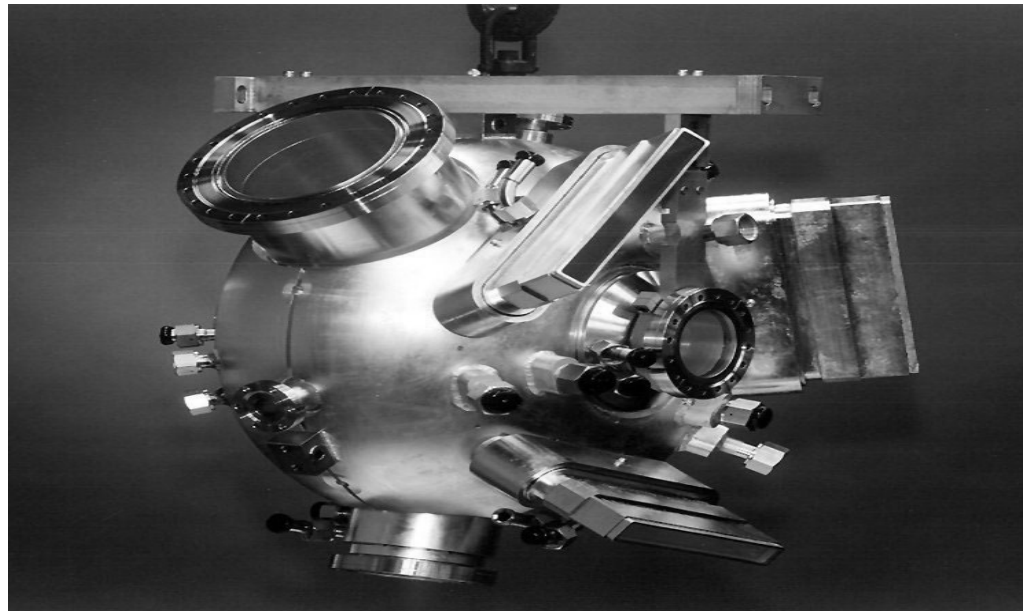


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11/06/2003

PEP-II RF Cavity Assemblies



CAV_13

PEP-II RF Cavity

8-19-97